	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{K}_{0^+}^{\#1\ \dagger}$	0	0
$\mathcal{K}_{0^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{K}_{1^+}^{\#1\ \alpha\beta}$	$\mathcal{K}_{1^+}^{\#2\ \alpha\beta}$	$\mathcal{K}_{1^-}^{\#1\ \alpha}$	$\mathcal{K}_{1^-}^{\#2\ \alpha}$
$\mathcal{K}_{1^+}^{\#1\ \dagger\alpha\beta}$	0	0	0	0
$\mathcal{K}_{1^+}^{\#2\ \dagger\alpha\beta}$	0	0	0	0
$\mathcal{K}_{1^-}^{\#1\ \dagger\alpha}$	0	0	$-k^2\ {}^{(4)}\kappa_2$	$\frac{Y_1 k^2}{2\sqrt{2}}$
$\mathcal{K}_{1^-}^{\#2\ \dagger\alpha}$	0	0	$\frac{Y_1 k^2}{2\sqrt{2}}$	$\frac{Y_2 k^2}{4}$

	$\mathcal{K}_{2^+}^{\#1\ \alpha\beta}$	$\mathcal{K}_{2^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{K}_{2^+}^{\#1\ \dagger\alpha\beta}$	$\frac{3 Y_3 k^2}{2}$	0
$\mathcal{K}_{2^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{J}_{0^+}^{\#1\ \dagger}$	0	0
$\mathcal{J}_{0^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

	$\mathcal{J}_{1^+}^{\#1\ \alpha\beta}$	$\mathcal{J}_{1^+}^{\#2\ \alpha\beta}$	$\mathcal{J}_{1^-}^{\#1\ \alpha}$	$\mathcal{J}_{1^-}^{\#2\ \alpha}$
$\mathcal{J}_{1^+}^{\#1\ \dagger\alpha\beta}$	0	0	0	0
$\mathcal{J}_{1^+}^{\#2\ \dagger\alpha\beta}$	0	0	0	0
$\mathcal{J}_{1^-}^{\#1\ \dagger\alpha}$	0	0	$\frac{Y_2 k^2}{4\,\mathrm{Det}(1^-)}$	$-\frac{Y_1 k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)}$
$\mathcal{J}_{1^-}^{\#2\ \dagger\alpha}$	0	0	$-\frac{Y_1 k^2}{2\sqrt{2}\,\mathrm{Det}(1^-)}$	$-\frac{k^2\ {}^{(4)}\kappa_2}{\mathrm{Det}(1^-)}$

	$\mathcal{J}_{2^+}^{\#1\ \alpha\beta}$	$\mathcal{J}_{2^-}^{\#1\ \alpha\beta\chi}$
$\mathcal{J}_{2^+}^{\#1\ \dagger\alpha\beta}$	$\frac{1}{\mathrm{Det}(2^+)}$	0
$\mathcal{J}_{2^-}^{\#1\ \dagger\alpha\beta\chi}$	0	0

Abbreviations used in matrices

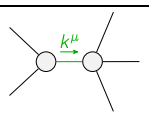
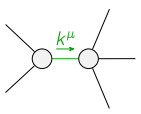
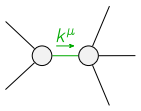
$$Y_1 == 2\ {}^{(4)}\kappa_2 + {}^{(4)}\kappa_7 \;\&\&\; Y_2 == 3\ {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 - 2\ {}^{(4)}\kappa_7 \;\&\&\; Y_3 == {}^{(4)}\kappa_1 + {}^{(4)}\kappa_2 \;\&\&$$

$$\mathrm{Det}(1^-) == -\frac{1}{8} k^4 (6\ {}^{(4)}\kappa_2 ({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2) + {}^{(4)}\kappa_7{}^2) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2} k^2 ({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)$$

Lagrangian

$$\begin{aligned}&{}^{(4)}\kappa_1\, \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + {}^{(4)}\kappa_2\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta} \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} - \frac{3}{2}\, {}^{(4)}\kappa_1\, \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} - \\ &\frac{3}{2}\, {}^{(4)}\kappa_2\, \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} - \frac{3}{2}\, {}^{(4)}\kappa_1\, \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \frac{3}{2}\, {}^{(4)}\kappa_2\, \partial_\beta \mathcal{K}^{\alpha\beta\chi} \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} + {}^{(4)}\kappa_7\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta}\end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$	
Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}_{0^-}^{\#1\ \alpha\beta\chi} == 0$	1	$\partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} == \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta}$
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta \mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}_{1^+}^{\#2\ \alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\chi\alpha\beta} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta}$
$\mathcal{J}_{1^+}^{\#1\ \alpha\beta} == 0$	3	$\partial_\delta \partial_\chi \partial^\alpha \mathcal{J}^{\chi\beta\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\beta\alpha\chi} == \partial_\delta \partial_\chi \partial^\beta \mathcal{J}^{\chi\alpha\delta} + \partial_\delta \partial^\delta \partial_\chi \mathcal{J}^{\alpha\beta\chi}$
$\mathcal{J}_{2^-}^{\#1\ \alpha\beta\chi} == 0$	5	$3\,\partial_\epsilon \partial_\delta \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon \partial^\epsilon \partial^\chi \partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon \partial_\delta \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon \partial^\epsilon \partial^\chi \partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon \partial^\epsilon \partial_\delta \partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi \partial^\phi \partial_\epsilon \partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	13	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{18\ {}^{(4)}\kappa_1{}^2 + 36\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_2 + 18\ {}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(6\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_2 + 6\ {}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)}$
	1	0	$-\frac{1}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(6\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_2 + 6\ {}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)} (30\ {}^{(4)}\kappa_2{}^2 + 6\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7 - {}^{(4)}\kappa_7{}^2 + 6\ {}^{(4)}\kappa_1 (5\ {}^{(4)}\kappa_2 + {}^{(4)}\kappa_7) + \sqrt{(9\ {}^{(4)}\kappa_1{}^4 + 1269\ {}^{(4)}\kappa_2{}^4 - 24\ {}^{(4)}\kappa_1{}^3\ {}^{(4)}\kappa_7 + 528\ {}^{(4)}\kappa_2{}^3\ {}^{(4)}\kappa_7 - 16\ {}^{(4)}\kappa_2{}^2\ {}^{(4)}\kappa_7{}^2 - 20\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7{}^3 + 3\ {}^{(4)}\kappa_7{}^4 + 2\ {}^{(4)}\kappa_1{}^2 (621\ {}^{(4)}\kappa_2{}^2 + 240\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7 + 34\ {}^{(4)}\kappa_7{}^2) + 4\ {}^{(4)}\kappa_1 (630\ {}^{(4)}\kappa_2{}^3 + 258\ {}^{(4)}\kappa_2{}^2\ {}^{(4)}\kappa_7 + 13\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7{}^2 - 5\ {}^{(4)}\kappa_7{}^3)))}$
	1	0	$\frac{1}{({}^{(4)}\kappa_1 + {}^{(4)}\kappa_2)(6\ {}^{(4)}\kappa_1\ {}^{(4)}\kappa_2 + 6\ {}^{(4)}\kappa_2{}^2 + {}^{(4)}\kappa_7{}^2)} (-30\ {}^{(4)}\kappa_2{}^2 - 6\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7 + {}^{(4)}\kappa_7{}^2 - 6\ {}^{(4)}\kappa_1 (5\ {}^{(4)}\kappa_2 + {}^{(4)}\kappa_7) + \sqrt{(9\ {}^{(4)}\kappa_1{}^4 + 1269\ {}^{(4)}\kappa_2{}^4 - 24\ {}^{(4)}\kappa_1{}^3\ {}^{(4)}\kappa_7 + 528\ {}^{(4)}\kappa_2{}^3\ {}^{(4)}\kappa_7 - 16\ {}^{(4)}\kappa_2{}^2\ {}^{(4)}\kappa_7{}^2 - 20\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7{}^3 + 3\ {}^{(4)}\kappa_7{}^4 + 2\ {}^{(4)}\kappa_1{}^2 (621\ {}^{(4)}\kappa_2{}^2 + 240\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7 + 34\ {}^{(4)}\kappa_7{}^2) + 4\ {}^{(4)}\kappa_1 (630\ {}^{(4)}\kappa_2{}^3 + 258\ {}^{(4)}\kappa_2{}^2\ {}^{(4)}\kappa_7 + 13\ {}^{(4)}\kappa_2\ {}^{(4)}\kappa_7{}^2 - 5\ {}^{(4)}\kappa_7{}^3)))}$
Resolved unitarity condition(s):		(Demonstrably impossible)	